MARKET BASKET ANALYSIS

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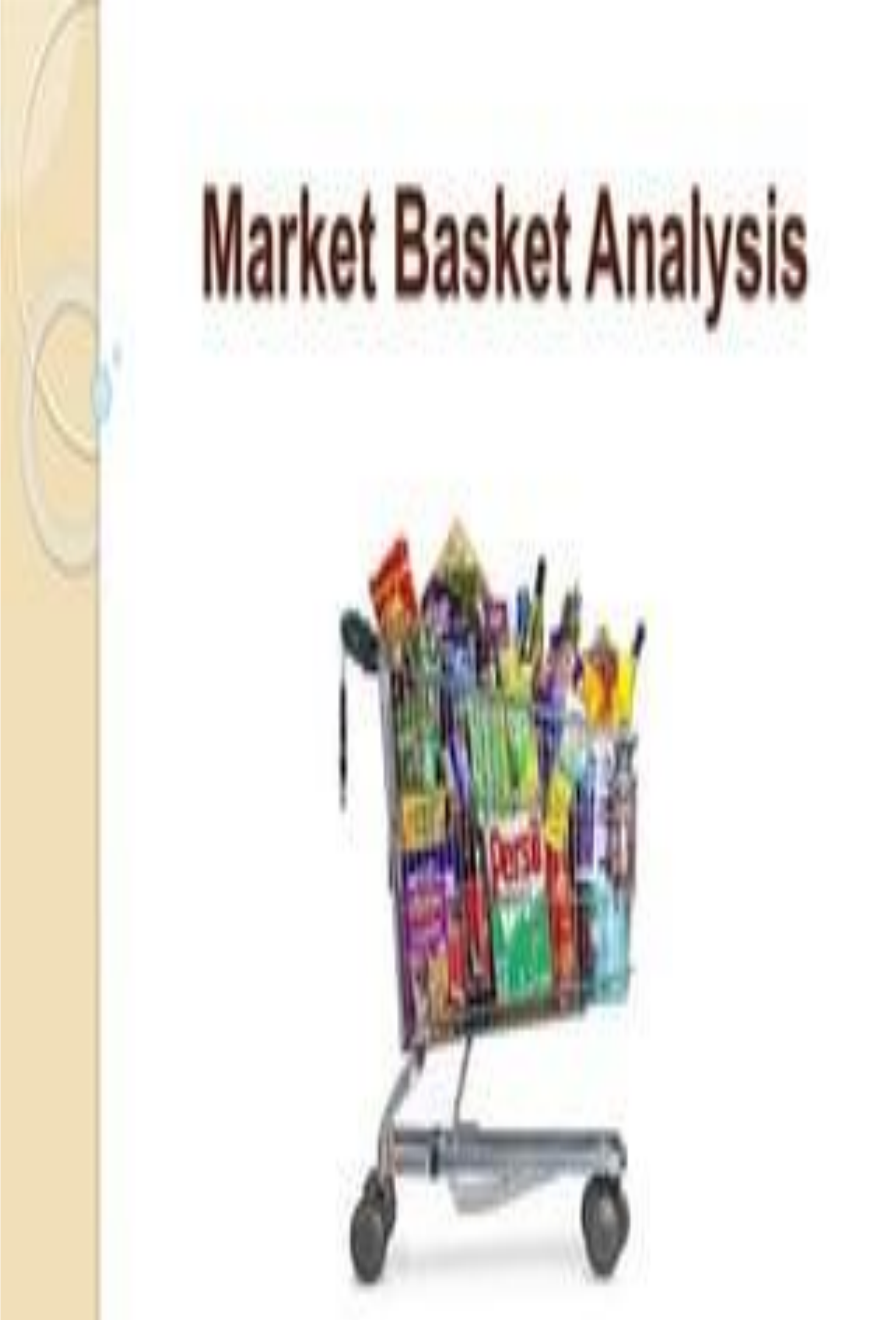


AGENDA

Introduction

Process

Dataset Description

Analysis

Findings and Recommendations

Business Benefits and Use Case-Specific Activities

MARKET BASKET ANALYSIS

What is Market Basket Analysis?

- A technique to find patterns in transactional data.
- Identifies items frequently purchased together.
- Example: Bread and butter are often bought as a pair.

Why is It Important?

- **Understand Customer Behavior**: Learn what customers prefer.
- **Boost Sales**: Optimize layouts and promotions.
- **Cross-Selling Opportunities**: Suggest complementary products.

Real-World Applications

- **Amazon**: "Frequently Bought Together" recommendations.

PROCESS FOR MARKET BASKET ANALYSIS

1. Gather transactional data, including purchase history, shopping carts, or invoices.
2. Analyze product sales and trends.
3. Use algorithms like Apriori to discover frequent item sets and generate association rules.
4. Interpret the discovered association rules to gain actionable insights.
5. Develop strategies based on the insights gained from the analysis.

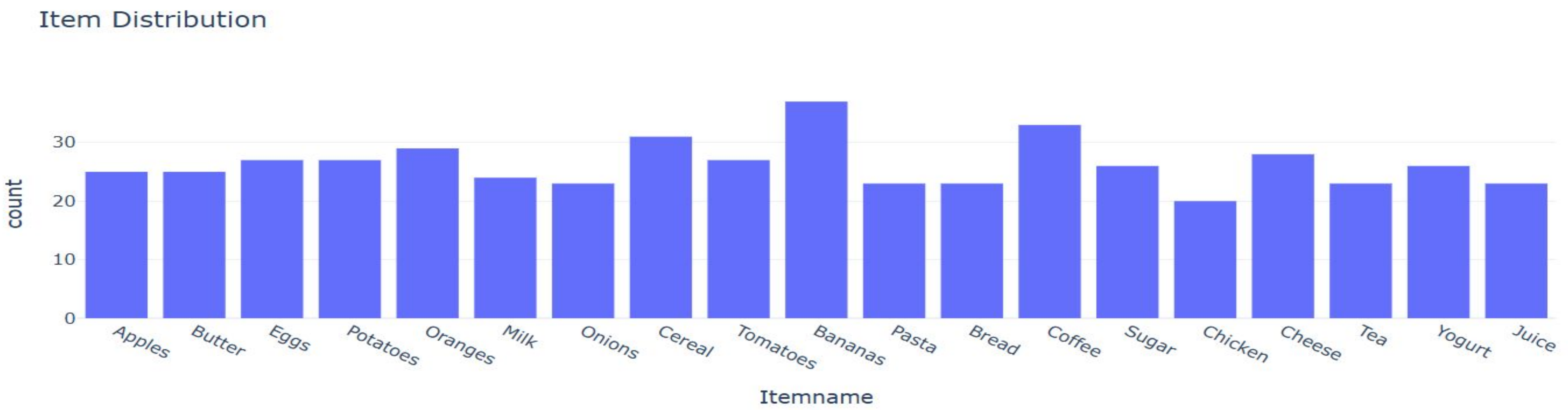
DATASET DESCRIPTION

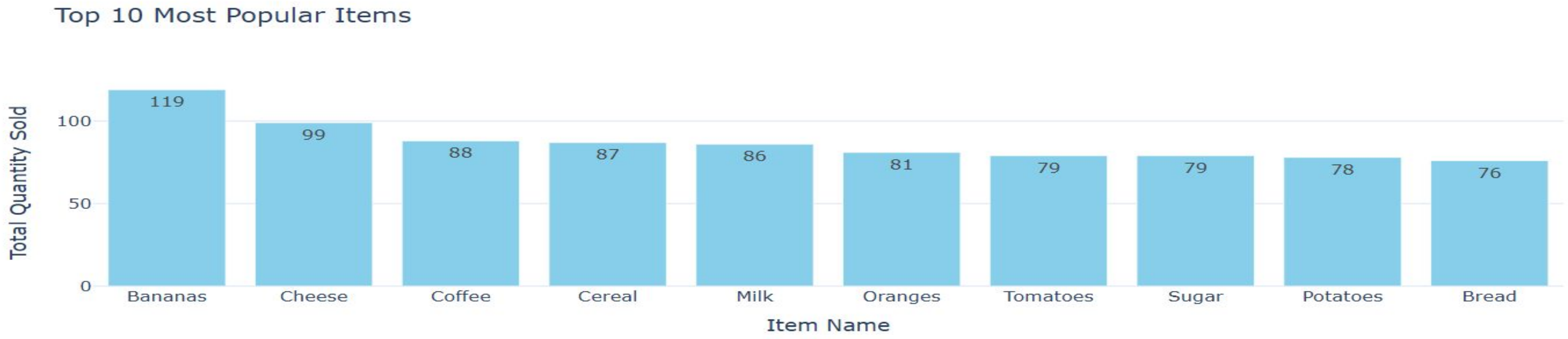
```python
data = pd.read_csv("market_basket_dataset.csv")
print(data.head())
```

	BillNo	Itemname	Quantity	Price	CustomerID
0	1000	Apples	5	8.30	52299
1	1000	Butter	4	6.06	11752
2	1000	Eggs	4	2.66	16415
3	1000	Potatoes	4	8.10	22889
4	1004	Oranges	2	7.26	52255

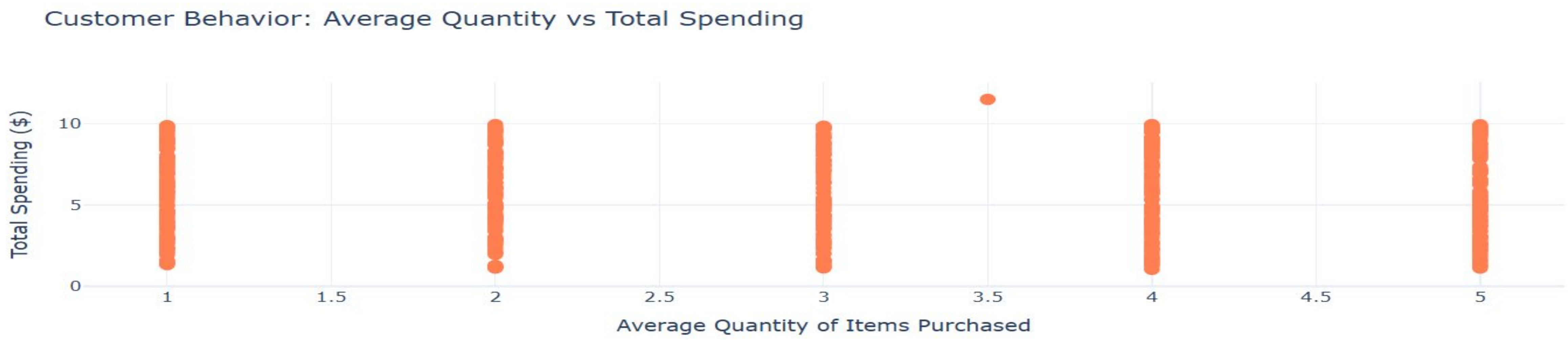
- **Data Types**: Numbers (Quantity, Price, Customer) & Text (Item name).
- **Unique Items**: Includes Apples, Butter, Eggs, etc.
- **Summary Stats**: Avg. quantity per transaction: 2.98.
- Avg. price: $5.62.

ANALYSIS


Item Distribution
count
0
10
20
30
Apples
Butter
Eggs
Potatoes
Oranges
Milk
Onions
Cereal
Tomatoes
Bananas
Pasta
Bread
Coffee
Sugar
Chicken
Cheese
Tea
Yogurt
Juice
Itemname


Top 10 Most Popular Items
Total Quantity Sold
0
50
100
119
99
88
87
86
81
79
79
78
76
Bananas
Cheese
Coffee
Cereal
Milk
Oranges
Tomatoes
Sugar
Potatoes
Bread
Item Name

ANALYSIS



Customer Behavior

CustomerID	Average Quantity	Total Spending
10504	1	2.04
10588	5	5.5
10826	1	5.67
11113	3	8.84
11267	1	8.87
11373	2	6.69
11430	3	4.85
11644	5	4.67

Total Spending
10
5
0
1 1.5 2 2.5 3 3.5 4 4.5 5
Average Quantity

APRIORI ALGORITHM

Data Organization & Transformation: Group items by transactions and convert them into a binary table format (1 for purchased, 0 for not purchased).

Frequent Item Set Discovery: Use the Apriori algorithm to find item sets that appear together often (e.g., "bread" and "butter").

Association Rules & Insights: Create rules to predict which items are likely to be bought together, using metrics like lift to evaluate the strength of these relationships.

FINDINGS

.Definitions:

1. **Antecedents**: Items customers buy first.
2. **Consequents**: Items bought alongside the antecedents.

.Key Metrics:

1. **Support**: Frequency of item pairs in transactions (e.g., Bread & Apples bought together in 4.58% of transactions).
2. **Confidence**: Likelihood of the consequent being bought after the antecedent (e.g., 30.43% chance of buying Apples if Bread is bought).
3. **Lift**: Measures the strength of the relationship between items (e.g., a lift of 1.86 indicates a strong positive relationship between Bread and Apples).

	antecedents	consequents	support	confidence	lift
0	(Bread)	(Apples)	0.045752	0.304348	1.862609
1	(Apples)	(Bread)	0.045752	0.280000	1.862609
2	(Butter)	(Apples)	0.026144	0.160000	0.979200
3	(Apples)	(Butter)	0.026144	0.160000	0.979200
4	(Cereal)	(Apples)	0.019608	0.096774	0.592258
5	(Apples)	(Cereal)	0.019608	0.120000	0.592258
6	(Cheese)	(Apples)	0.039216	0.214286	1.311429
7	(Apples)	(Cheese)	0.039216	0.240000	1.311429
8	(Chicken)	(Apples)	0.032680	0.250000	1.530000
9	(Apples)	(Chicken)	0.032680	0.200000	1.530000

RECOMMENDATION

Low Lift Items

- Market them separately
- Individual promotions.

High Lift items

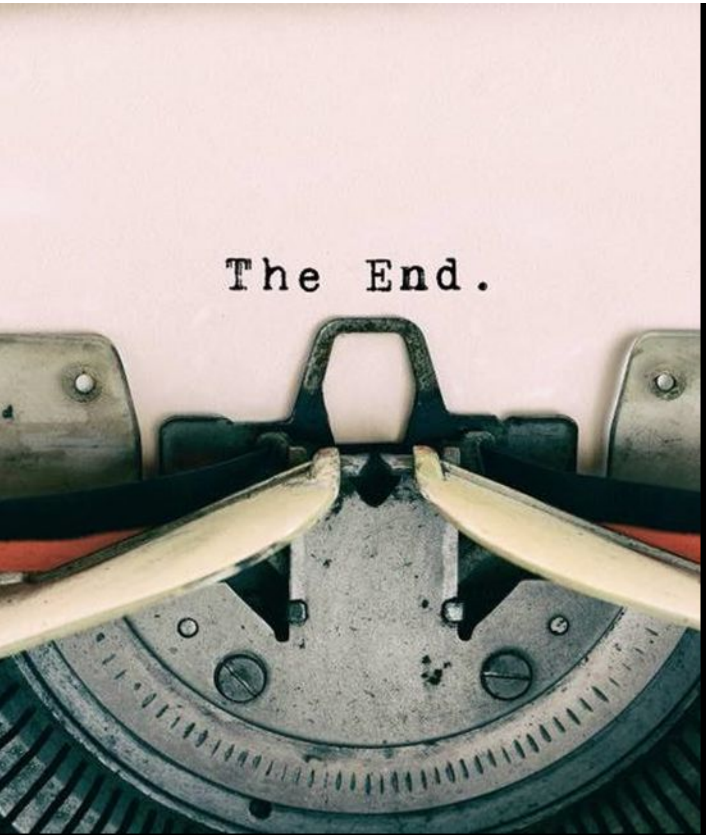
- Special discounts when customers buy both
- Product Placement:
- Bundle Offers
- Cross-Selling

BUSINESS BENEFITS AND USE CASE-SPECIFIC ACTIVITIES

Increase Revenue through Product Bundling and Cross-Selling: Leverage associations to bundle products and offer discounts to drive higher sales.

Enhance Customer Retention with Targeted Promotions: Build customer loyalty by offering product pairings and discounts that match their buying behavior.

Optimize Product Placement for Higher Conversion Rates: Use insights to place items together in-store and online, improving visibility and purchase likelihood.



THANK YOU